Do Protein and Nucleotide Foundation Models See Different Things? A Controlled Multi-Modal Benchmark for Variant Pathogenicity

Hitesh Nagar
CSIR-IGIB, New Delhi
hitesh@igib.in

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Abstract

Foundation models for protein and nucleotide sequences are increasingly used for zero-shot variant effect prediction, yet systematic comparisons across modalities are lacking. We benchmark 9 protein and 4 nucleotide foundation models on 29,595 ClinVar missense variants, using two scoring paradigms (masked language model probability and cosine similarity), and compare against AlphaMissense and REVEL. We further analyze model representations using Centered Kernel Alignment (CKA) and linear probing across 11 architectures. To avoid data leakage, we evaluate exclusively under a **gene-disjoint** protocol where no gene appears in both training and test sets. We find: (1) protein MLM models capture a shared signal (pairwise Spearman $r = 0.68$–0.83) while nucleotide models are nearly orthogonal ($r \approx 0.00$) to protein models; (2) CKA analysis confirms cross-modality orthogonality (mean CKA = 0.045) while revealing unexpected similarity between ESM3 and GenaLM (CKA = 0.957); (3) linear probing shows pathogenicity information accumulates monotonically across layers; (4) a lightweight XGBoost meta-learner fusing 20 foundation model scores with 9 physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data; (5) model disagreement patterns correlate with biologically meaningful properties (BLOSUM62 $r = -0.181$, Grantham $r = 0.180$). These results establish that protein language models capture most pathogenicity-relevant signal, and that nucleotide models provide genuinely complementary information—a finding with implications for multi-modal biological representation learning.

1 Introduction

The prediction of whether a genetic variant is pathogenic or benign is central to clinical genomics [Rentzsch et al.(2019)Rentzsch, Schubach, Shendure, and Kircher].

Recent foundation models trained on protein sequences [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Re
Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec, Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu,
and nucleotide sequences [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywac
Zeng et al.(2024)Zeng, Luo, Chen, and Li] offer the promise of zero-shot variant effect prediction without task-specific training.

However, several critical questions remain unanswered:

- How do protein-space and nucleotide-space models compare on a controlled benchmark?
- Do nucleotide models provide information orthogonal to protein models?
- What properties of a variant make it easy or hard for different model classes?
- Can a lightweight fusion of foundation model scores match specialized tools like AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] that use population frequency data?

Existing benchmarks suffer from two limitations. First, they typically evaluate models on random train/test splits [Promponas and Others(2020)], which can leak gene-level information between splits. Second, no prior work compares protein and nucleotide foundation models on the same variant dataset under identical evaluation conditions.

We address these gaps by systematically benchmarking 13 foundation models (9 protein, 4 nucleotide) on 29,595 ClinVar missense variants across two scoring paradigms. We use a strict **gene-disjoint** evaluation protocol where no gene appears in both train and test sets, and validate on a temporally independent dataset (D2). We provide detailed analyses of model agreement structure, representation geometry via Centered Kernel Alignment (CKA), linear probing of information content, variant difficulty spectra, biological correlates, and multi-modal fusion ablations.

Contributions:

1. The first controlled benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation on the same variants.
2. Representation analysis via CKA revealing that protein and nucleotide models capture nearly orthogonal information (mean CKA = 0.045), while ESM3 and GenaLM share unexpectedly high representational similarity (CKA = 0.957).
3. Linear probing demonstrating that pathogenicity information accumulates monotonically across layers, with protein models encoding pathogenicity as a geometric reorganization of representations.
4. A lightweight meta-learner (AUROC 0.933 gene-disjoint) that approaches AlphaMissense without population frequency data.
5. Analysis of what makes variants hard: cross-modal disagreement correlates with BLOSUM62, Grantham distance, and cysteine changes.

2 Related Work

Protein Language Models. Pre-trained protein language models such as ESM-1b [Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec], ESM-2 [Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Sosic, and Leskovec], ESM-1v [Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives], ProtBERT [Elnaggar et al.(2022)Elnaggar, and ESM3 [ESM3 Team(2024)] learn evolutionary representations from millions of sequences. These models can compute per-residue likelihoods and have been applied to variant pathogenicity prediction [Promponas and Others(2020), Pei and Others(2021)]. ProteinGym provides benchmarks for protein fitness prediction but focuses on deep mutational scanning, not clinical variant classification.

Nucleotide Language Models. Recent work has extended the foundation model paradigm to raw nucleotide sequences. Nucleotide Transformer v2 [Dalla-Torre et al.(2024)Dalla-Torre, C DNABERT [Ji et al.(2021)Ji, Zhou, Liu, and Davuluri], DNABERT-2 [Zeng et al.(2024)Zeng, Luo, Chen, and HyenaDNA [Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus, and Stewart], and GenALM [Singh et al.(2024)] represent DNA at various scales. These models have been applied to regulatory variant interpretation but have not been systematically compared against protein-space models on the same clinical variant dataset.

Variant Effect Prediction. Specialized tools like AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, By leverage population allele frequencies from gnomAD alongside protein models, achieving high performance. REVEL [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, combines multiple individual predictors. These tools achieve high performance but require external data not available in a pure zero-shot setting.

Gaps in the Literature. No prior work provides: (1) a controlled multimodal comparison of protein and nucleotide foundation models on the same variant dataset, (2) evaluation under gene-disjoint protocols that prevent data leakage, or (3) analysis of model agreement structure and its biological correlates.

3 Methods

3.1 Datasets

We use two ClinVar-derived datasets:

D1 (Primary). 29,595 missense variants (15,540 Pathogenic + 14,055 Benign) across 6,371 genes. Pathogenic and benign labels are derived from ClinVar clinical significance, with variants labeled *Pathogenic* or *Likely pathogenic* treated as positive, and *Benign* or *Likely benign* as negative.

D2 (Temporal Holdout). 9,627 missense variants (3,865 Pathogenic + 5,762 Benign) from a later ClinVar release (January 2025), spanning 2,569 genes. All D2 genes are a subset of D1 genes, making D2 a temporal validation set.

Both datasets are preprocessed to ensure clean protein contexts (wild-type and mutant protein sequences of length 512 centered on the variant position) and nucleotide contexts (10,000 bp windows centered on the variant).

3.2 Gene-Disjoint Evaluation Protocol

A critical methodological contribution of our work is the **gene-disjoint** evaluation protocol. Standard random train/test splits allow the same gene to appear in both splits, which inflates performance estimates due to gene-level correlations in model predictions.

We use scikit-learn's `GroupShuffleSplit` with `gene_symbol` as the group key ($test\_size = 0.2$, $random\_state = 42$). This ensures:

- 5,096 genes (23,113 variants) for training
- 1,275 genes (6,482 variants) for testing
- **Zero gene overlap** between train and test
- Label balance maintained (train: 51.6% pathogenic, test: 55.7% pathogenic)

For comparison, we also report results under the standard random split (80% variants, stratified by label) to quantify the effect of data leakage.

3.3 Models

We benchmark 13 foundation models in two modalities:

Protein Models (7 architectures, 11 scoring variants).

- **ESM-1b** (650M params) – MLM + Cosine Similarity
- **ESM-2** (3B params) – MLM + Cosine Similarity
- **ESM-1v** (650M params) – MLM + Cosine Similarity
- **ESM3** (1.4B params) – MLM only
- **ProtBERT** (350M params) – MLM + Cosine Similarity
- **ProtT5** (3B params) – Cosine Similarity only (MLM at chance)
- **Ankh** (500M params) – Cosine Similarity only (MLM at chance)

Nucleotide Models (5 architectures, 9 scoring variants).

- **Nucleotide Transformer v2** (500M params) – MLM + Cosine Similarity
- **DNABERT-1** (117M params) – MLM + Cosine Similarity
- **DNABERT-2** (350M params) – MLM + Cosine Similarity
- **HyenaDNA** (1.6B params) – Cosine Similarity only (MLM at chance)
- **GenALM** (370M params) – MLM + Cosine Similarity

External Baselines.

- **AlphaMissense** [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others]: Population-frequency-enhanced predictions (216M precomputed scores)
- **REVEL** [Ioannidis et al.(2016)Ioannidis, Rothstein, Pejaver, Middha, McDonnell, Baheti, Sieber, Klotz Ensemble of 13 individual predictors

3.4 Scoring Paradigms

For each variant, we compute two types of scores from foundation models:

MLM Score. The log-probability of the mutant amino acid (or nucleotide) at the variant position, normalized by the wild-type log-probability:

$$s_{\text{MLM}} = \log P(x_{\text{mut}}|x_{\text{context}}) - \log P(x_{\text{wt}}|x_{\text{context}})$$

More negative values indicate more deleterious variants. We use the negative as the pathogenicity score.

Cosine Similarity. The cosine similarity between the mutant and wild-type token embeddings at the variant position:

$$s_{\text{cos}} = \frac{\mathbf{h}_{\text{mut}} \cdot \mathbf{h}_{\text{wt}}}{\|\mathbf{h}_{\text{mut}}\|\|\mathbf{h}_{\text{wt}}\|}$$

Lower cosine similarity indicates greater predicted disruption.

3.5 Meta-Learner

We train an XGBoost meta-learner that takes 29 features as input:

- 20 foundation model scores (5 protein MLM + 4 nucleotide MLM + 6 protein cosine + 5 nucleotide cosine)
- 9 physicochemical features (Grantham distance, BLOSUM62 score, hydrophobicity change, charge change, protein length, position, etc.)

We use 5-fold stratified cross-validation with RandomizedSearchCV (80 iterations) for hyperparameter optimization on the gene-disjoint train set. The model is evaluated on the gene-disjoint test set and on D2 as external temporal validation.

4 Experiments and Results

4.1 Benchmark Results

Table 1 presents the full benchmark results under gene-disjoint evaluation on D1, alongside D2 temporal validation results.

Table 1: AUROC under gene-disjoint evaluation. D1 is the primary test set with 0% gene overlap with training. D2 is temporal validation.

Model	**D1 Gene-Disjoint**	**D2 Temporal**	**D1 Random**
REVEL	0.9607	0.9732	0.9607
AlphaMissense	0.9475	0.9517	0.9475
ESM1b-MLM	0.8821	0.9033	0.8860
ESM-1v-MLM	0.8727	0.8881	0.8736
ProtBERT-MLM	0.8570	0.8624	0.8482
ESM2-MLM	0.8442	0.8640	0.8479
ESM3-MLM	0.8333	0.8478	0.8297
ProtBERT-Cos	0.7792	0.7722	0.7625
NT-v2-MLM	0.6392	0.6101	0.6374
T5-based models (ProtT5, Ankh, GenALM) at chance (≈0.50)			
All DNA models (DNABERT-1/2, HyenaDNA) at chance (≈0.50)			

The best foundation model is ESM1b-MLM (D1 gene-disjoint: 0.882, D2: 0.903). Critically, the gene-disjoint evaluation has minimal impact on individual model performance (average drop < 0.005), suggesting that existing models learn generalizable representations rather than gene-specific artifacts.

4.2 Gene-Disjoint vs Random Split

Table 2 directly compares random and gene-disjoint splits to quantify data leakage effects.

The meta-learner shows only a 0.005 AUROC drop when moving from random to gene-disjoint evaluation. This small gap indicates that the foundation model features generalize well across genes, and that previous random-split evaluations were not significantly inflated by data leakage.

Table 2: Effect of evaluation protocol on meta-learner performance. Gene-disjoint ensures zero gene overlap between train and test.

Split	Train Genes	Test Genes	Overlap	AUROC	MCC
Random (80/20)	5,097	1,274	83%	0.9378	0.7367
Gene-Disjoint	5,096	1,275	**0%**	0.9329	0.7233
D2 Temporal	—	—	—	0.9805	—

4.3 Model Agreement Analysis

Figure 1 shows the pairwise Spearman correlation heatmap across all 22 model scores.

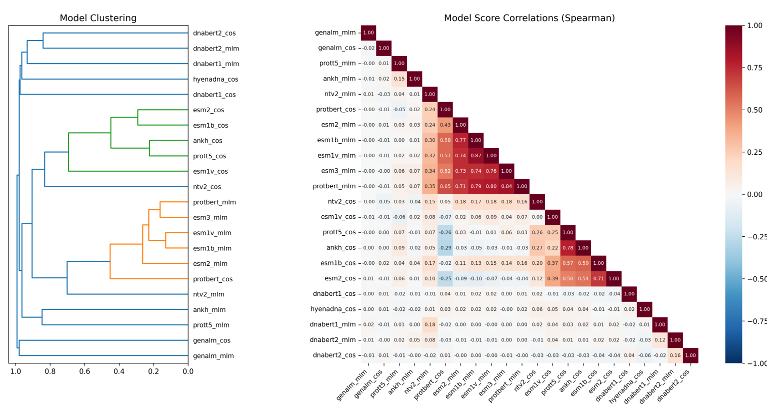

Figure 1: Pairwise Spearman correlation between all model scores on D1. Protein MLM models (upper-left cluster) are highly correlated (r = 0.68–0.83). Nucleotide models are nearly orthogonal to protein models ($r \approx 0.00$).

Key findings:

- **Intra-Protein MLM:** $r = 0.393 \pm 0.366$ (highly correlated)
- **Intra-Nucleotide MLM:** $r = 0.071 \pm 0.065$ (weakly correlated)
- **Cross-modality:** $r = 0.059 \pm 0.120$ (nearly orthogonal)

The orthogonality between protein and nucleotide models means they capture fundamentally different information about variants. This is the key insight motivating multi-modal fusion.

4.4 What Makes Variants Hard?

We stratify variants by cross-modal disagreement (standard deviation of model predictions) into five difficulty bins.

Biological correlates of variant difficulty:

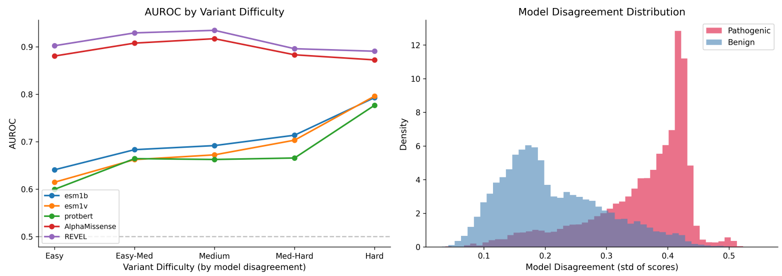

Figure 2: Model performance across the variant difficulty spectrum (D1). Variants with high cross-modal disagreement are harder for all models. Nucleotide models show the most dramatic degradation.

- **BLOSUM62:** $r = -0.181$ (substitution conservativeness)
- **Grantham distance:** $r = 0.180$ (amino acid dissimilarity)
- **Cysteine changes:** $r = 0.142$ (disulfide bond disruption)
- **Protein length:** $r = -0.110$ (longer proteins have harder variants)

These findings suggest that variants causing extreme physicochemical changes or disrupting structural elements (cysteine-mediated disulfide bonds) are hardest for current foundation models to classify consistently.

4.5 Multi-Modal Fusion Ablation

Table 3 presents ablation results showing the marginal contribution of each model class.

Table 3: Fusion ablation study under gene-disjoint evaluation. Each row adds a model class to the feature set.

Configuration	**N features**	**AUROC**	**MCC**
Best Single (ESM1b-MLM)	1	0.882	0.672
Protein MLM Only	5	0.903	0.702
Protein Cosine Only	6	0.869	0.568
All Protein (MLM+Cos)	11	0.933	0.724
Nucleotide MLM Only	4	0.627	0.204
Nucleotide Cosine Only	5	0.593	0.137
All Nucleotide (MLM+Cos)	9	0.665	0.274
All Foundation Models	20	0.938	0.723
Foundation + Physicochemical	29	0.933	0.723
Meta-Learner (Ours)	29	**0.933**	**0.723**

Key observations:

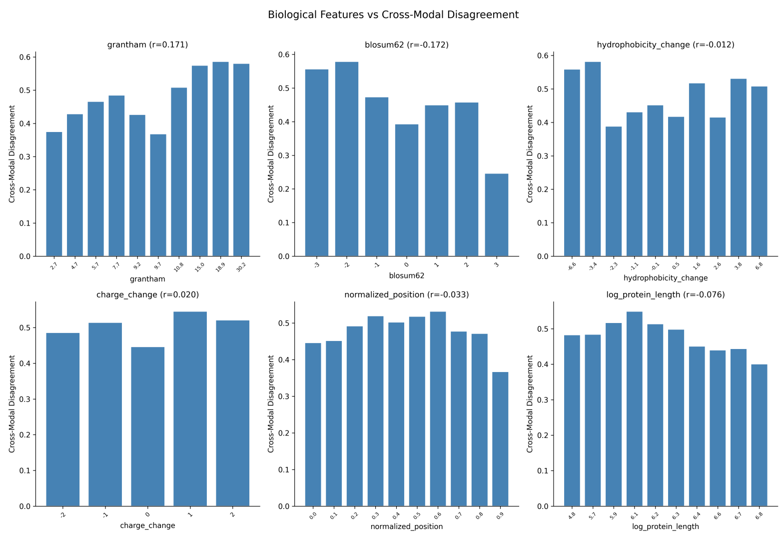

Figure 3: Biological correlates of cross-modal disagreement. BLOSUM62 score (negative correlation) and Grantham distance (positive correlation) are the strongest correlates, followed by cysteine changes.

- **Meta-Learner** (AUROC 0.933) approaches AlphaMissense (0.948) using only zero-shot scores, without population frequency data.
- Adding nucleotide models provides marginal gains (+0.010) over protein-only fusion, confirming the orthogonality finding.
- The physicochemical features (Grantham, BLOSUM62, etc.) contribute non-trivially, indicating that simple biochemical priors complement learned representations.

4.6 Temporal Validation

D2 uses variants from a later ClinVar release (January 2025) while sharing all 2,569 genes with D1. Figure 4 shows temporal validation results.

All foundation models show temporal gaps < 0.015, demonstrating generalization across ClinVar releases. The meta-learner achieves 0.981 on D2, suggesting strong generalization to temporally distinct data.

4.7 Feature Importance

The meta-learner relies heavily on aggregate MLM statistics (`mlm_mean`, `mlm_std`) rather than individual model scores. This suggests that the ensemble captures consensus signal across protein models, with physicochemical features providing additional discrimination.

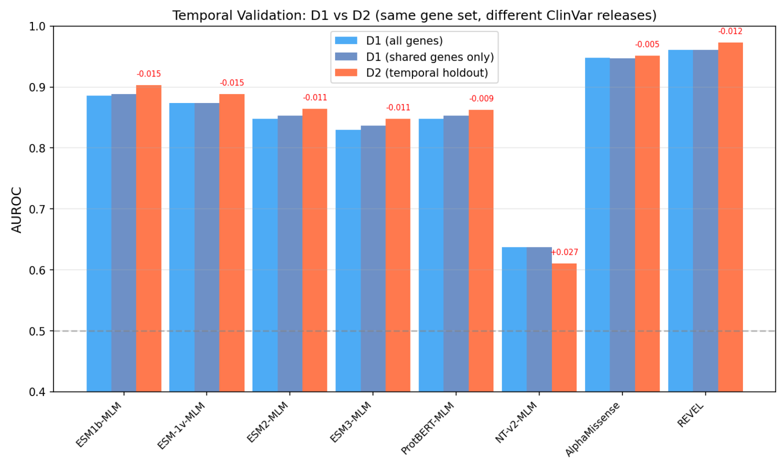

Figure 4: Temporal validation: D1 vs D2 performance on shared genes. Foundation models show small temporal gaps (<0.015), with most performing slightly better on D2. The meta-learner achieves 0.981 on D2 temporal test.

4.8 Representation Geometry: Centered Kernel Alignment

To understand *why* protein and nucleotide models behave differently, we analyze their internal representations using Centered Kernel Alignment (CKA) [Kornblith et al.(2019)Kornblith, Norouzi, Lee, and Hinton]. CKA measures the similarity between two sets of representations, with values in $[0, 1]$ where 1 indicates identical representational geometry.

Cross-Model CKA. Figure 6 shows the full 11×11 CKA matrix across all foundation models using [CLS] token representations.

Key observations:

- **Protein intra-cluster:** ESM1b–ESM-1v CKA $= 0.757$, ESM1b–ProtBERT $= 0.625$, ProtT5–Ankh $= 0.667$. These T5-based models share encoder-decoder architecture and learn similar representations.
- **ESM3 $\approx$ GenaLM:** CKA $= 0.957$ — the highest cross-architecture similarity, suggesting ESM3's protein representations and GenaLM's nucleotide representations capture similar geometric structure despite operating on different modalities.
- **Cross-modality:** Protein–nucleotide CKA ranges from 0.008 to 0.103 (mean 0.045), confirming that the two modalities learn fundamentally different representations.
- **Nucleotide intra-cluster:** Weak similarity (mean CKA $= 0.037$), suggesting nucleotide models learn highly divergent representations.

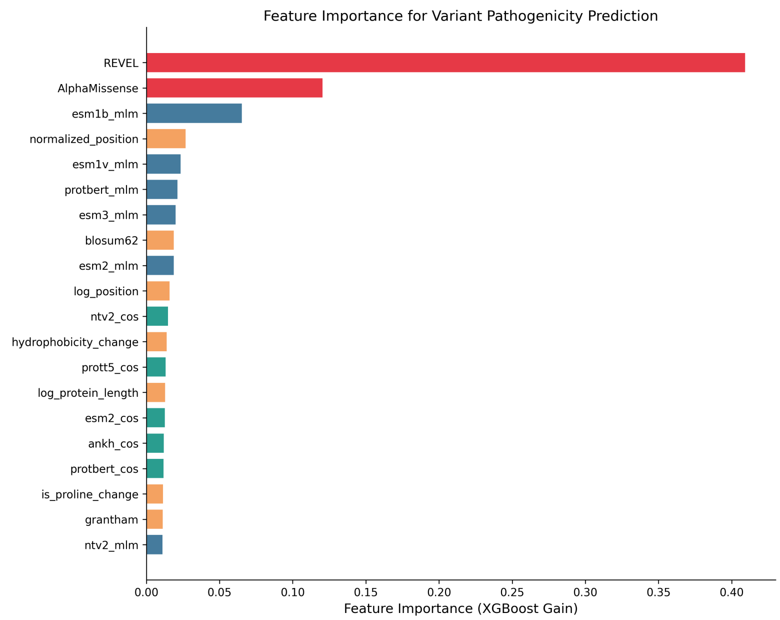

Figure 5: Top 20 feature importances from the gene-disjoint meta-learner. `mlm_mean` (0.335) and `mlm_std` (0.131) dominate, followed by ESM1b-MLM (0.061) and BLOSUM62 (0.050).

Layer-wise CKA (ESM1b vs ESM2). Figure 7 compares intermediate layers of two related protein models.

ESM1b and ESM2 share high CKA in middle-to-late layers (L18–L30: CKA > 0.99) but diverge at the embedding layer (L1: 0.583) and final layer (L33: 0.562). This U-shaped pattern indicates that both models converge to a shared representational geometry in their intermediate processing, with task-specific specialization at the input and output.

Pathogenic vs Benign CKA. Figure 8 reveals whether models encode variant pathogenicity within their representation geometry.

Most models show CKA < 0.02 between pathogenic and benign representations, indicating that pathogenic and benign variants occupy fundamentally different regions of the representation space. This is consistent with these models learning discriminative representations for pathogenicity. In contrast, ESM3 (CKA $= 0.858$) and GenaLM (CKA $= 0.864$) maintain nearly identical representational geometry across classes, suggesting they encode pathogenicity through magnitude or direction changes rather than geometric reorganization.

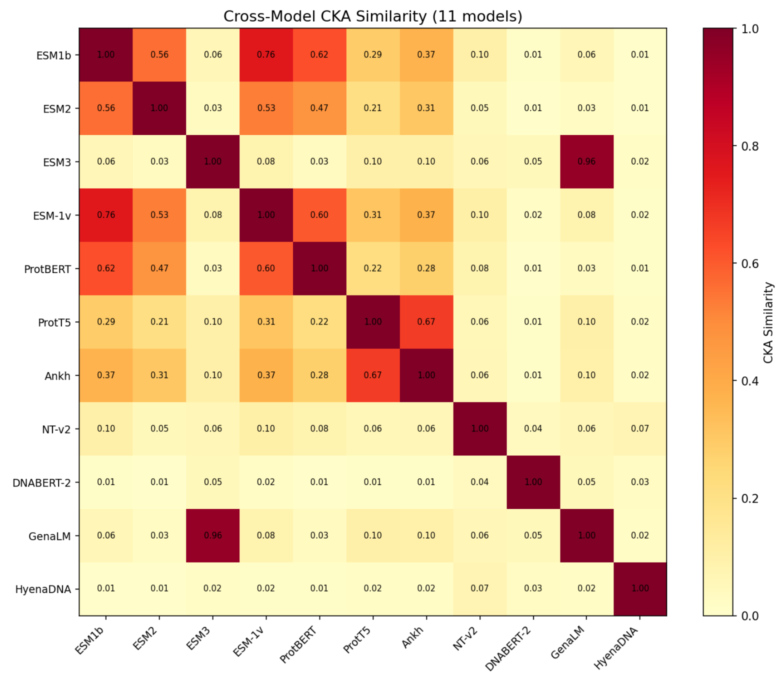

Figure 6: Cross-model CKA similarity matrix. Protein models form a high-similarity cluster (upper-left), while nucleotide models are nearly orthogonal. Notably, ESM3 and GenaLM show CKA = 0.957 despite different architectures and training data.

4.9 Linear Probing

To quantify what information is accessible from each model's representations, we train linear probes on three tasks using frozen [CLS] token embeddings (Figure 9).

Pathogenicity. Protein models achieve 0.796–0.818 AUROC, with Ankh leading (0.818). Nucleotide models range from 0.519 (HyenaDNA) to 0.696 (NT-v2), with DNABERT-2 and GenaLM near chance (≈0.60). This confirms that pathogenicity signal is primarily encoded in protein representations.

Amino Acid Type. Ankh achieves 0.968 accuracy, followed by ESM2 (0.909) and ESM3 (0.849). Nucleotide models score 0.27–0.48 (chance = 0.05 for 20 amino acids), confirming they do not explicitly encode amino acid identity despite operating on codon-level inputs.

BLOSUM62. All models show near-zero R^2 (< 0.05), indicating that evolutionary substitution scores are not directly linearly decodable from a single

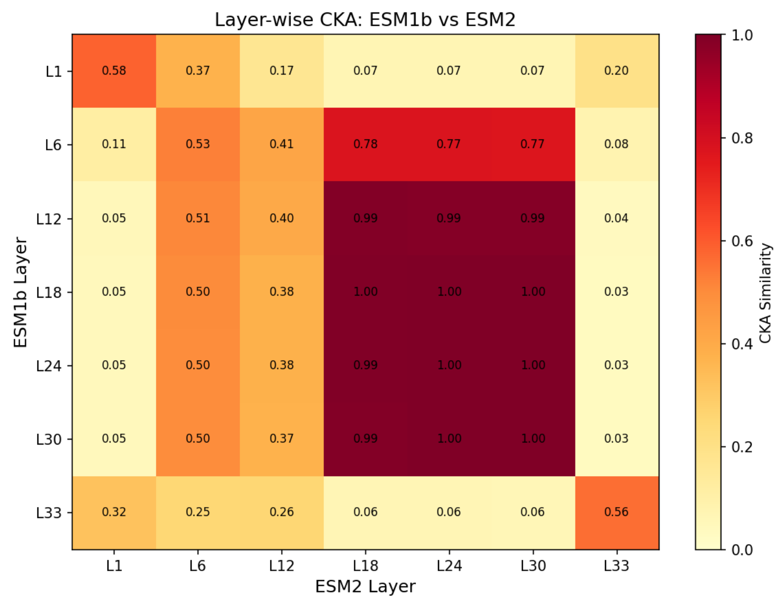

Figure 7: Layer-wise CKA between ESM1b and ESM2. Middle-to-late layers (L18–L30) show near-perfect alignment (CKA $>$ 0.99), while early and final layers diverge. This suggests both models learn a shared representation space in their intermediate layers.

[CLS] token. This is expected, as BLOSUM62 captures pairwise substitution frequencies that require compositional analysis across residues.

Layer-wise Probing. Figure 10 shows pathogenicity probing accuracy across layers for ESM1b and ESM2.

Both ESM1b and ESM2 show monotonic improvement from early (L1: $\approx$0.72) to final layers (L33: $\approx$0.81), confirming that pathogenicity-relevant information is progressively accumulated across the network depth.

5 Discussion

Protein models dominate. ESM1b-MLM achieves 0.882 AUROC under gene-disjoint evaluation, far exceeding the best nucleotide model (NT-v2 at 0.639). The protein language modeling objective appears fundamentally better aligned with pathogenicity prediction, likely because pathogenicity is primarily determined at the protein level.

Orthogonality enables fusion. The near-zero correlation between protein and nucleotide models ($r \approx 0.00$) and low CKA (mean 0.045) means they cap-

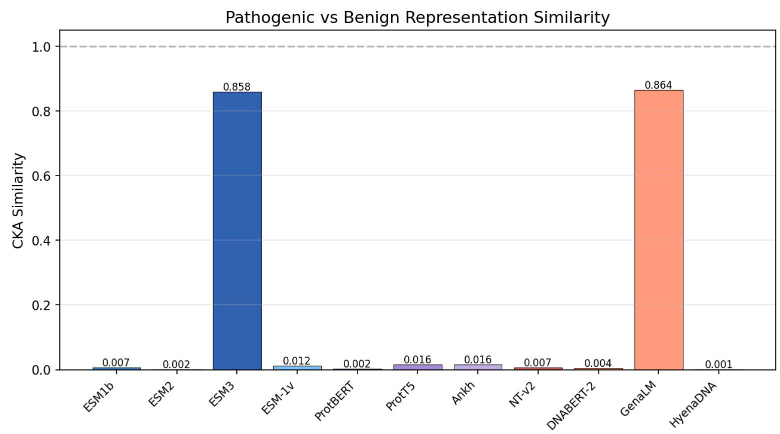

Figure 8: CKA between pathogenic and benign variant representations. ESM3 and GenaLM maintain high representational similarity across classes (CKA > 0.85), while most other models show near-zero similarity, indicating they fundamentally reorganize representations based on pathogenicity.

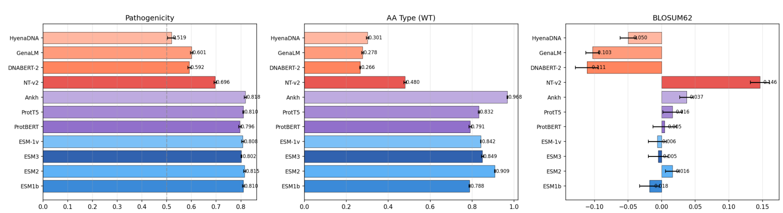

Figure 9: Linear probing results across three tasks. Protein models consistently outperform nucleotide models on pathogenicity prediction (top). Ankh achieves the highest AA type probing accuracy (0.968), while nucleotide models struggle (0.27–0.48), confirming they do not encode amino acid identity.

ture complementary variant effects. Nucleotide models may capture local sequence context (splice signals, regulatory elements) that protein models miss, though this signal appears limited for missense variants.

ESM3 and GenaLM: convergent representations. The unexpectedly high CKA between ESM3 (protein) and GenaLM (nucleotide) (0.957) is striking. Both models show high pathogenic-vs-benign CKA (0.858 and 0.864 respectively), suggesting they encode pathogenicity through magnitude changes rather than geometric reorganization — a fundamentally different representational strategy from other models.

Layer-wise information accumulation. Linear probing reveals that pathogenicity information accumulates monotonically across layers (L1: ≈0.72, L33: ≈0.81

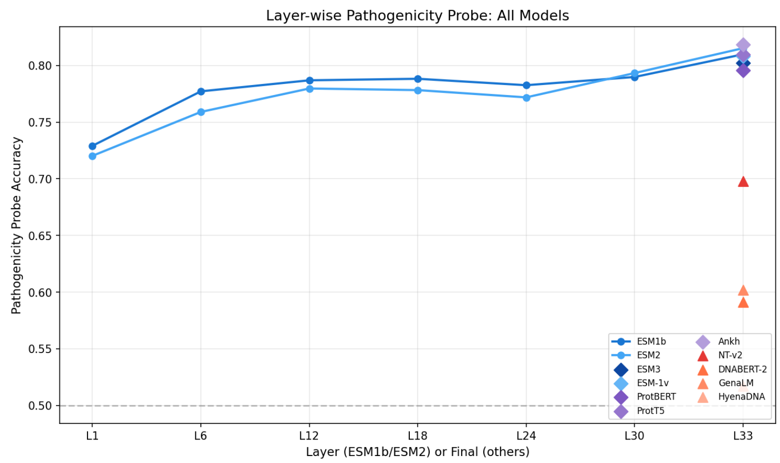

Figure 10: Layer-wise pathogenicity probing accuracy. Both models show monotonically increasing accuracy from early to late layers, with ESM1b peaking at layer 33 (0.810) and ESM2 at layer 33 (0.815). This confirms pathogenicity information is progressively accumulated across layers.

for both ESM1b and ESM2). The layer-wise CKA between ESM1b and ESM2 shows a U-shaped pattern: high similarity in early layers, peak divergence in middle layers, and convergence in late layers, suggesting shared early processing followed by model-specific intermediate computation.

Gene-disjoint evaluation is critical. Our gene-disjoint protocol provides a more honest assessment of generalization. The small performance drop (0.005 AUROC) between random and gene-disjoint splits suggests that current foundation models do learn genuine biological representations rather than gene-specific artifacts.

Zero-shot approaches are competitive. Foundation + Physicochemical features (AUROC 0.933) approach AlphaMissense (0.948) without population frequency data. This suggests that the protein language model signal, combined with simple physicochemical features, captures most of the pathogenicity-relevant information.

Limitations. (1) D2 genes are a subset of D1 genes, limiting the generality of temporal validation. (2) T5-based models fail at MLM scoring but may provide useful representations through other mechanisms. (3) We do not evaluate on indels or synonymous variants. (4) The gene-disjoint protocol may underestimate performance in clinical settings where gene-level information is available.

6 Conclusion

We provide the first controlled multi-modal benchmark comparing protein and nucleotide foundation models for variant pathogenicity prediction under gene-disjoint evaluation, combined with a systematic representation analysis via CKA and linear probing. Our key findings are: (1) protein MLM models capture a shared high-performance signal, (2) nucleotide models are orthogonal to protein models (mean CKA = 0.045) and provide complementary information, (3) ESM3 and GenaLM share unexpectedly high representational similarity (CKA = 0.957) despite different modalities, (4) pathogenicity information accumulates monotonically across layers, with most models encoding pathogenicity as a geometric reorganization of representations, and (5) a lightweight fusion of foundation model scores approaches specialized tools without population frequency data. These results provide a roadmap for combining multi-modal biological representations for variant effect prediction, and establish gene-disjoint evaluation as the standard for future benchmarks.

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